DEPENDANCY INJECTION

Example 1: Using field

```java
package com.example.DependencyInjection;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class DependencyInjectionApplication {

    public static void main(String[] args) {

        ApplicationContext context=SpringApplication.run(DependencyInjectionApplication.class, args);
        Employee e=context.getBean(Employee.class);
        e.display();
    }
}
```

```java
package com.example.DependencyInjection;

import org.springframework.stereotype.Component;

@Component  1 usage
public class Company {
    public void display_details()  1 usage
    {
        System.out.println("Company: ZOHO");
    }
}
```

```java
package com.example.DependencyInjection;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component  2 usages
public class Employee {
    int id=101;  1 usage
    String name="Suji";  1 usage

    @Autowired
    private Company Company; //field injection

    public void display()  1 usage
    {
        System.out.println("ID: "+id);
        System.out.println("Name: "+name);
        Company.display_details();
    }
}
```

```
2026-10-15T19:15:50.520+05:30  INFO 31528 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionApplic
2026-10-15T19:15:50.523+05:30  INFO 31528 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionApplic
2026-10-15T19:15:50.963+05:30  INFO 31528 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionApplic
ID: 101
Name: Suji
Company: ZOHO

Process finished with exit code 0
```

Example 2: Using constructor

```java
package com.example.DependencyInjection;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component  2 usages
public class Employee {
    int id=101;  1 usage
    String name="Suji";  1 usage

    @Autowired
    public Employee(Company company) {
        Company = company;
    }
    private Company Company; //field injection  2 usages

    public void display()  1 usage
    {
        System.out.println("ID: "+id);
        System.out.println("Name: "+name);
        Company.display_details();
    }
}
```

```
2026-10-15T20:06:12.828+05:30  INFO 34416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
2026-10-15T20:06:12.831+05:30  INFO 34416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
2026-10-15T20:06:13.314+05:30  INFO 34416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
ID: 101
Name: Suji
Company: ZOHO

Process finished with exit code 0
```

Example 3: Using setter

```java
package com.example.DependencyInjection;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component  2 usages
public class Employee {
    int id=101;  1 usage
    String name="Suji";  1 usage

    @Autowired
    public void setCompany(Company company) {
        Company = company;
    }

    private Company Company; //field injection  2 usages

    public void display()  1 usage
    {
        System.out.println("ID: "+id);
        System.out.println("Name: "+name);
        Company.display_details();
    }
}
```

```
2026-10-15T20:07:34.598+05:30  INFO 22416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
2026-10-15T20:07:34.601+05:30  INFO 22416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
2026-10-15T20:07:35.038+05:30  INFO 22416 --- [DependencyInjection] [           main] c.e.D.DependencyInjectionAppli
ID: 101
Name: Suji
Company: ZOHO

Process finished with exit code 0
```